

ENGINE LEVEL

is the geometry right?

○ Line-of-sight engine

profile sampling · curvature + refraction

5 switchable defects, injected one at a time

○ Benchmark

15 terrain cases, expectations derived

from the critical distance, not intuition

4 behavioural properties, metamorphic

○ Mutation analysis

the whole benchmark, run against

each deliberately broken engine

● 4 defects killed · 1 equivalent

equivalence proven, then confirmed

by a paired sweep

DESIGN LEVEL

is the inference right?

○ Synthetic landscape

anisotropic ridges · a lake flooded to a constant, exactly as real DEMs do

○ Ground truth by construction

S0 exchangeable → p exactly uniform

S1 best of K → known theoretical

power, ceiling 85 %

○ Paired contrasts

correct · water unmasked · tight crop

same data — only the defect differs

● Calibration · power · mechanism

each defect measured against an

exact expectation, not an assumption

DIAGNOSTICS · ANY REAL STUDY

D1 null placements touching water = 0

D2 orientation coverage = 100 %

Only a validated engine and a diagnosed design touch real terrain.